2021 Summer Qual

III.

(a) $$\frac{\partial\left((y-X\beta)^T(y-X\beta)+\lambda\beta^T\beta\right)}{\partial\beta}$$

$$= -2X^T(y-X\beta)+2\lambda\beta = 0$$

$$\Rightarrow X^Ty - X^TX\beta - \lambda\beta = 0$$

$$\Rightarrow (X^TX+\lambda I)\beta = X^Ty$$

$$\Rightarrow \hat{\beta} = (X^TX+\lambda I)^{-1}X^Ty$$

$$= \left(I+\lambda(X^TX)^{-1}\right)^{-1}\hat{\beta}$$

$\blacksquare$

(b) $$LOCV = \frac{1}{n}PRESS = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_{(-i)}\right)^2$$

$$\hat{\beta}^R = (X^TX+\lambda I_p)^{-1}X^Ty$$

$$\hat{\beta}^R_{(i)} = (X_{(i)}^TX_{(i)}+\lambda I_p)^{-1}X_{(i)}^Ty_{(i)} = \left(X^TX+\lambda I_p - x_ix_i^T\right)^{-1}(X^Ty - x_iy_i)$$

$$= \left\{(X^TX+\lambda I_p)^{-1} + \frac{(X^TX+\lambda I_p)^{-1}x_ix_i^T(X^TX+\lambda I_p)^{-1}}{1-x_i^T(X^TX+\lambda I_p)^{-1}x_i}\right\}(X^Ty - x_iy_i)$$

$$= \hat{\beta}^R - (X^TX+\lambda I_p)^{-1}x_iy_i + \frac{(X^TX+\lambda I_p)^{-1}x_ix_i^T(X^TX+\lambda I_p)^{-1}(X^Ty - x_iy_i)}{1-x_i^T(X^TX+\lambda I_p)^{-1}x_i}$$

$$= \hat{\beta}^R - \frac{(X^TX+\lambda I_p)^{-1} x_i (y_i - \hat{y}_i(\lambda))}{1 - x_i^T (X^TX+\lambda I_p)^{-1} x_i}$$

$$y_i - \hat{y}_{(-i)} = y_i - x_i^T \left\{ \hat{\beta}^R - \frac{(X^TX+\lambda I_p)^{-1} x_i e_i(\lambda)}{1 - x_i^T (X^TX+\lambda I_p)^{-1} x_i} \right\}$$

$$= e_i + \frac{x_i^T (X^TX+\lambda I_p)^{-1} x_i e_i(\lambda)}{1 - x_i^T (X^TX+\lambda I_p)^{-1} x_i}$$

$$= \frac{e_i}{1 - \underbrace{x_i^T (X^TX+\lambda I_p)^{-1} x_i}_{= A_{ii}(\lambda)}}$$

$$LOCV = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{e_i(\lambda)}{1 - A_{ii}(\lambda)} \right)^2$$

$\boxtimes$

$\boxed{2}$.

(a) $\pi_1+\pi_2=1,\quad X \sim \begin{cases} F_1 & \text{w.p. } \pi_1 \\ F_2 & \text{w.p. } \pi_2 \end{cases}$

$$
\begin{aligned}
T(F) &= \int b(x)\,dF(x) \\
&= \int b(x)\,d\left[\pi_1 F_1(x) + \pi_2 F_2(x)\right] \\
&= \pi_1 \int b(x)\,dF_1(x) + \pi_2 \int b(x)\,dF_2(x) \\
&= \pi_1 T(F_1) + \pi_2 T(F_2)
\end{aligned}
$$

(b)

$$
\begin{aligned}
\int IF(x;T,F)\,dF_1 &= \lim_{t\downarrow 0} \frac{T((1-t)F+tF_1) - T(F)}{t} \\
&= \lim_{t\downarrow 0} \frac{\int b(x)\,d[(1-t)F+tF_1] - \int b(x)\,dF}{t} \\
&= \lim_{t\downarrow 0} \frac{t\int b(x)\,dF + t\int b(x)\,dF_1}{t} \\
&= T(F) - T(F_1) \\
&= \pi_1 T(F_1) + \pi_2 T(F_2) - T(F_1) \\
&= \pi_2 T(F_1) - \pi_2 T(F_2)
\end{aligned}
$$

$$
\Rightarrow T(F_1) - T(F_2) = \frac{1}{\pi_2}\int IF(x;T,F)\,dF_1(x). \qquad \boxtimes
$$

(c) $T(F) = \int b(x)\, dF(x)$

$= \int \sin x\, dF(x)$

$IF(x; T, F) = \sin x - \int \sin x\, dF$

$|T(F)| = |\int \sin x\, dF| \leq \int |\sin x|\, dF \leq \int 1\, dF = 1$

$|IF(x; T, F)| = |\sin x - T(F)|$

$\leq 1 + 1 = 2$ $\therefore$ B- robust .

$\boxed{3}$

(a) $X^TX\hat{\theta} = X^Ty \xRightarrow{\text{BLUE}} \lambda^T\hat{\theta} = \lambda^T(X^TX)^- X^Ty$: linear

$E[\lambda^T(X^TX)^-X^Ty] = \lambda^T(X^TX)^-X^TX\theta$

$= \lambda^T\theta$; UE

Let $\tilde{\theta} = d^Ty$ s.t. $E\, d^Ty = d^TX\theta = \lambda^T\theta \Rightarrow \lambda^T = d^TX$

$0 \le Var((d^T - \lambda^T(X^TX)^-X^T)y)$

$= \sigma^2 (d^T - \lambda^T(X^TX)^-X^T)(d - X(X^TX)^-\lambda)$

$= \sigma^2(d^Td - \lambda^T(X^TX)^-\lambda - \cancel{\lambda^T(X^TX)^-\lambda} + \cancel{\lambda^T(X^TX)^-\lambda})$

$= Var(\tilde{\theta}) - Var(\lambda^T\hat{\theta})$

$\Rightarrow Var(\lambda^T\hat{\theta}) \le Var(\tilde{\theta})$

(b) $SSE = y^T(I - X(X^TX)^-X^T)y$

$df = N - rank(X) = 12 - 3 = 9$

$(X^TX)^-X^T(I - H_X) = 0$

$$T = \frac{\dfrac{\lambda^T\hat{\theta} - 2}{\sqrt{\lambda^T(X^TX)^-\lambda\sigma^2}}}{\sqrt{\dfrac{\hat{\sigma}^2}{\sigma^2}}} \perp \sim t(9)$$

$\Rightarrow T > t_{0.05,9}$

$\boxed{4}$.

(a) $\mathbb{E}(y_{ij} \mid u_i) = e^{\beta x_{ij} + u_i}$

$$\mathbb{E}(y_{ij}) = \mathbb{E}[\mathbb{E}(y_{ij} \mid u_i)] = \mathbb{E}(e^{\beta x_{ij} + u_i})$$

$$= e^{\beta x_{ij}} \mathbb{E} e^{u_i}, \quad u_i \sim N(0, \sigma^2)$$

$$= e^{\beta x_{ij} + \sigma^2/2}$$

$$Var(y_{ij}) = \mathbb{E}[\underbrace{Var(y_{ij} \mid u_i)}_{= e^{\beta x_{ij} + u_i}}] + \underbrace{Var(\mathbb{E}(y_{ij} \mid u_i))}_{Var(e^{\beta x_{ij} + u_i}) = e^{2\beta x_{ij}} Var(e^{u_i})}$$

$$= e^{\beta x_{ij} + \sigma^2/2} + e^{2\beta x_{ij}}(e^{2\sigma^2} - e^{\sigma^2})$$

$$= e^{2\beta x_{ij}}(\mathbb{E} e^{2u_i} - (\mathbb{E} e^{u_i})^2)$$

$$= e^{2\beta x_{ij}}(e^{2\sigma^2} - e^{\sigma^2})$$

Actually in Poisson, $\mathbb{E} = Var$

But $Var(Y_{ij}) > \mathbb{E} Y_{ij}$

$\Rightarrow$ Y_{ij} does not follow a poisson distribution. ☒

(b) complete data $\{y_{ij}, u_i\}$

$Y_{ij} \mid u_i \sim poisson(e^{\beta x_{ij} + u_i})$

$$\mathbb{P}(Y_{ij} = y_{ij} \mid u_i) = \frac{e^{-e^{\beta x_{ij} + u_i}} (e^{\beta x_{ij} + u_i})^{y_{ij}}}{y_{ij}!}$$

$$\log \mathbb{P}(Y_{ij} = y_{ij} \mid u_i) = y_{ij}(\beta x_{ij} + u_i) - e^{\beta x_{ij} + u_i} - \log y_{ij}!$$

$$u_i \sim N(0, \sigma^2)$$

$$\Rightarrow \log f(u_i) = -\frac{1}{2}\log 2\pi\sigma^2 - \frac{u_i^2}{2\sigma^2}$$

$$l_c(\beta) = \sum_{i=1}^{m}\sum_{j=1}^{n}\left[y_{ij}(\beta x_{ij} + u_i) - e^{\beta x_{ij} + u_i} - \log y_{ij}!\right]$$
$$- \sum_{i=1}^{m}\frac{u_i^2}{2\sigma^2} - \frac{m}{2}\log 2\pi\sigma^2$$

$$Q(\beta|\beta^{(t)}) = \mathbb{E}_{\beta^{(t)}}\left[l_c(\beta) \mid \{y_{ij}\}\right]$$

since $m_i^{(t)}(k) = \mathbb{E}_{\beta^{(t)}}[u_i^k \mid \{y_{ij}\}]$

$$M_i^{(t)}(s) = \mathbb{E}_{\beta^{(t)}}[e^{su_i} \mid \{y_{ij}\}]$$

$$\mathbb{E}_{\beta^{(t)}}(u_i|y) = m_i^{(t)}(1)$$

$$\mathbb{E}_{\beta^{(t)}}(u_i^2|y) = m_i^{(t)}(2)$$

$$\mathbb{E}_{\beta^{(t)}}(e^{u_i}|y) = M_i^{(t)}(1).$$

$$\Rightarrow Q(\beta|\beta^{(t)}) = \sum_{i=1}^{m}\sum_{j=1}^{n} y_{ij}\beta x_{ij} + \sum_{i=1}^{m}\sum_{j=1}^{n} y_{ij}\, m_i^{(t)}(1)$$
$$- \sum_{i=1}^{m}\sum_{j=1}^{n} e^{\beta x_{ij}} M_i^{(t)}(1)$$
$$- \sum_{i=1}^{m}\frac{m_i^{(t)}(2)}{2\sigma^2} - \frac{m}{2}\log 2\pi\sigma^2$$

dep only on β

$$\Rightarrow Q(\beta|\beta^{(t)}) = \beta\sum_{i=1}^{m}\sum_{j=1}^{n} x_{ij}y_{ij}$$
$$- \sum_{i=1}^{m}\sum_{j=1}^{n} M_i^{(t)}(1)e^{\beta x_{ij}} - \frac{m}{2}\log 2\pi\sigma^2$$

(c) $Q(\beta) = \beta \sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} y_{ij} - \sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) e^{\beta x_{ij}} - \frac{m}{2} \log 2\pi\sigma^2$

$Q'(\beta) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} y_{ij} - \sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) x_{ij} e^{\beta x_{ij}}$

$Q''(\beta) = - \sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) x_{ij}^2 e^{\beta x_{ij}} < 0$

$\Rightarrow \exists!$ maximizer in $\mathbb{R}$.

(d) $\beta^{(t+1)} = \arg\max Q(\beta \mid \beta^{(t)})$

$g_t(\beta) := Q'(\beta) = \sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} y_{ij} - \sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) x_{ij} e^{\beta x_{ij}}$

$g_t'(\beta) := Q''(\beta) = - \sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) x_{ij}^2 e^{\beta x_{ij}}$

Using Newton-Raphson,

$$\beta_{new} = \beta_{old} - \frac{g_t(\beta_{old})}{g_t'(\beta_{old})}$$

$$\Rightarrow \underbrace{\beta_{new}}_{= \beta^{(t+1)}} = \underbrace{\beta_{old}}_{= \beta^{(t)}} + \frac{\sum_{i=1}^{m} \sum_{j=1}^{n} x_{ij} y_{ij} - \sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) x_{ij} e^{\beta x_{ij}}}{\sum_{i=1}^{m} \sum_{j=1}^{n} M_i^{(t)}(1) x_{ij}^2 e^{\beta x_{ij}}}$$